# IGCSE 0607 Practice Questions — E10.7

## Scatter diagrams

**Total Questions:** 15 — **Total Marks:** 96 Marks

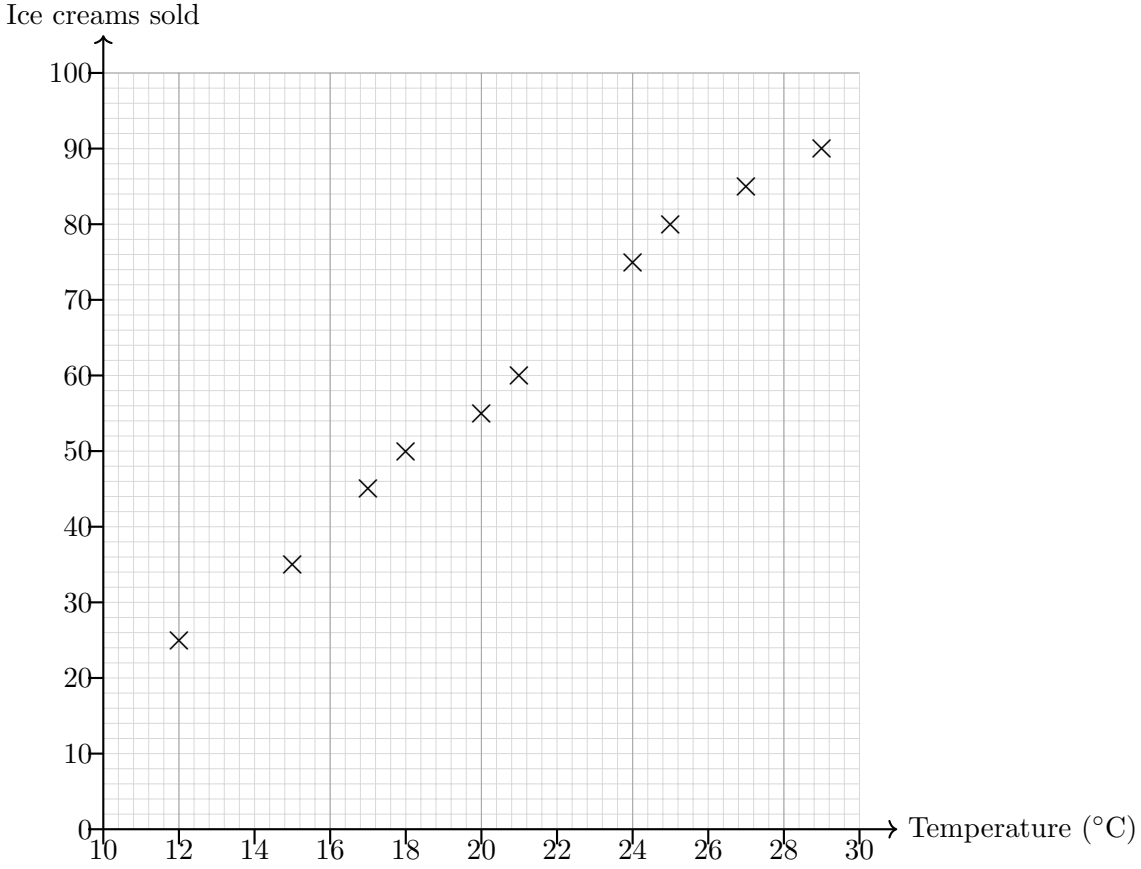
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**1** The scatter diagram shows the relationship between the maximum daily temperature, $x$°C, and the number of ice creams sold, $y$, on 10 different days.




**(a)** State the type of correlation shown in the scatter diagram.

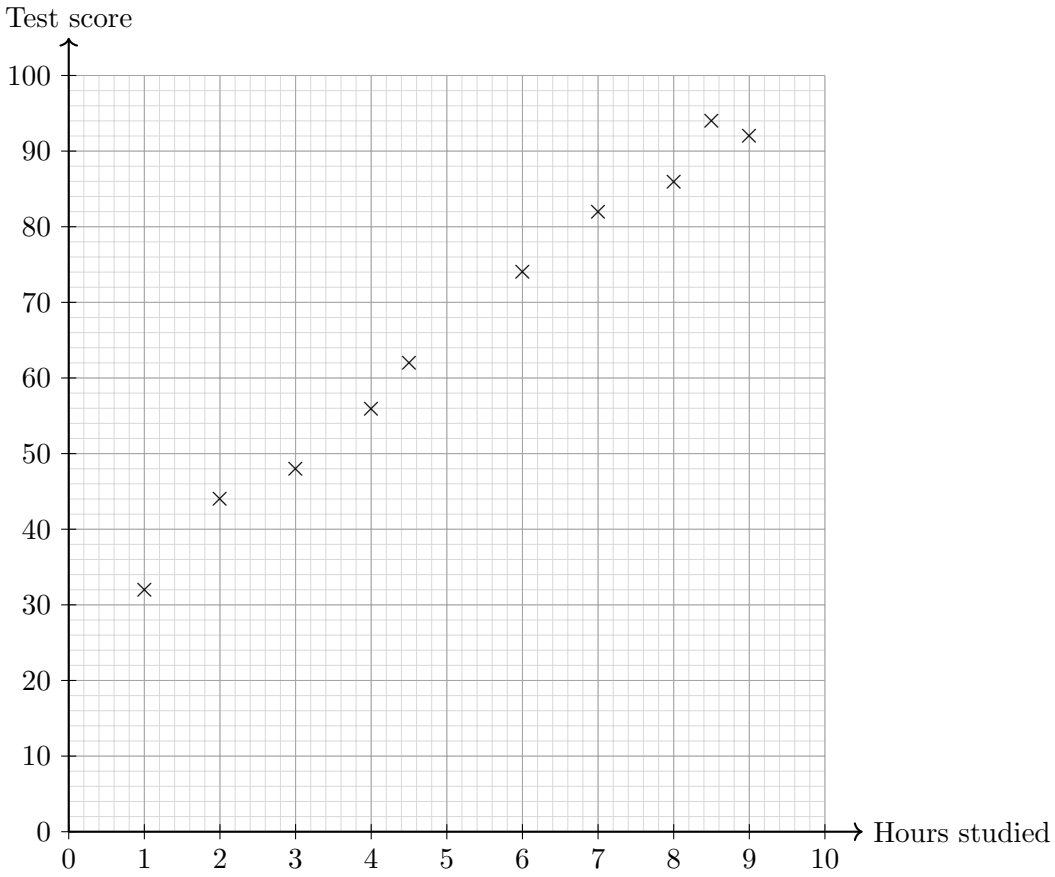
..... **[1]**

**(b)** On the scatter diagram, draw a line of best fit. **[1]**

**(c)** Use your line of best fit to estimate the number of ice creams sold on a day when the maximum temperature is 22°C.

..... **[1]**

**2** The scatter diagram shows the number of hours studied and the test score achieved by 10 students.




**(a)** What type of correlation is shown by the scatter diagram?

..... **[1]**

**(b)** On the scatter diagram, draw a line of best fit. **[1]**

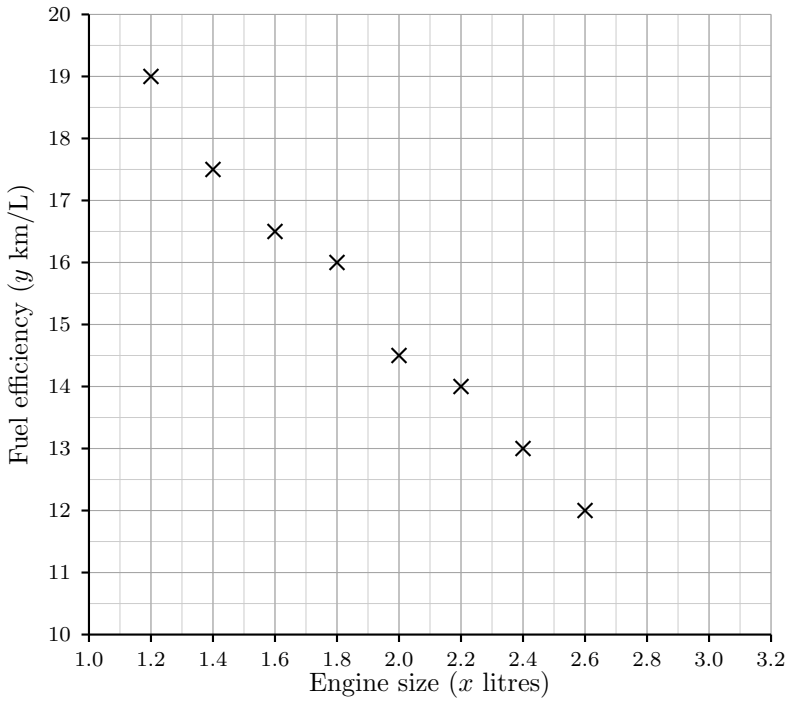
**(c)** A student studied for 5.5 hours.
Use your line of best fit to estimate their test score.

..... **[1]**

**3** A motoring magazine records the engine size, $x$ litres, and the fuel efficiency, $y$ km/L, for 10 cars. The table shows the results.

| Engine size ($x$ L) | 1.2 | 1.4 | 1.6 | 1.8 | 2.0 | 2.2 | 2.4 | 2.6 | 2.8 | 3.0 |
|---|---|---|---|---|---|---|---|---|---|---|
| Efficiency ($y$ km/L) | 19.0 | 17.5 | 16.5 | 16.0 | 14.5 | 14.0 | 13.0 | 12.0 | 11.5 | 11.0 |

**(a)** The first 8 points are plotted on the scatter diagram below.
Plot the remaining 2 points, $(2.8, 11.5)$ and $(3.0, 11.0)$, on the grid.




**[2]**

**(b)** State the type of correlation shown by the scatter diagram.

..... **[1]**

**(c)** **(i)** Calculate the mean engine size, $\bar{x}$, and the mean fuel efficiency, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

**(ii)** On the scatter diagram in part (a), plot the mean point $(\bar{x}, \bar{y})$ and draw a straight line of best fit by eye. **[2]**

**(d)** Use your line of best fit to estimate the fuel efficiency of a car with an engine size of 2.5 litres.

..... km/L **[1]**

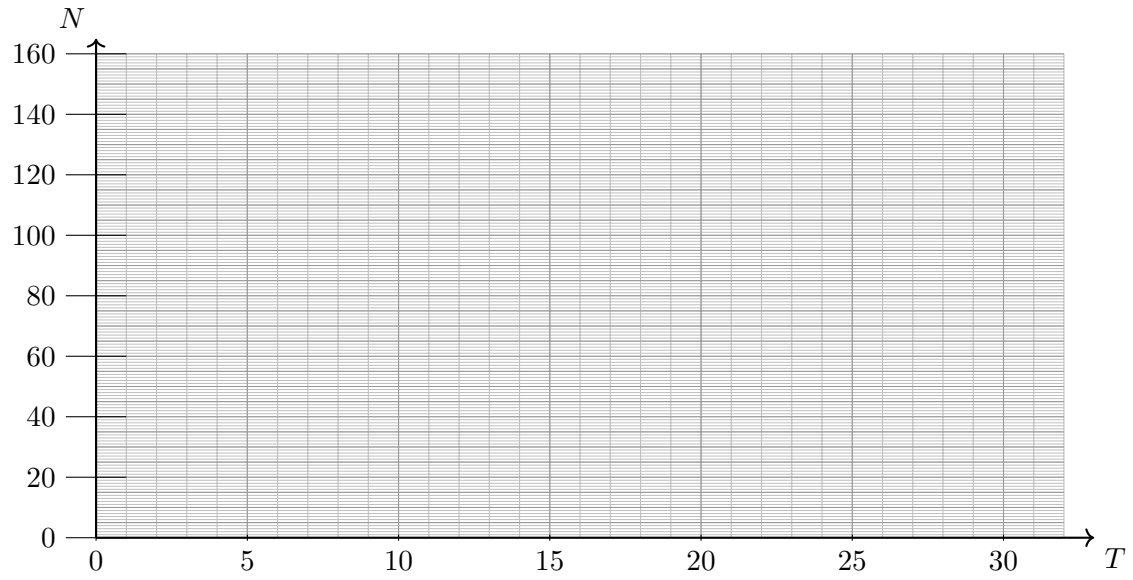
**(e)** Explain why your line of best fit should not be used to estimate the fuel efficiency of a car with an engine size of 5.0 litres.

..... **[1]**

**4** The table shows the outside temperature, $T$°C, and the number of hot drinks, $N$, sold by a cafe on 10 different days.

| Temperature ($T$°C) | 5 | 8 | 12 | 15 | 18 | 20 | 22 | 25 | 28 | 30 |
|---|---|---|---|---|---|---|---|---|---|---|
| Number of hot drinks ($N$) | 140 | 125 | 100 | 95 | 70 | 65 | 50 | 40 | 25 | 15 |

**(a)** Complete the scatter diagram.
The first four points have been plotted for you.




**[2]**

**(b)** What type of correlation is shown by the scatter diagram?

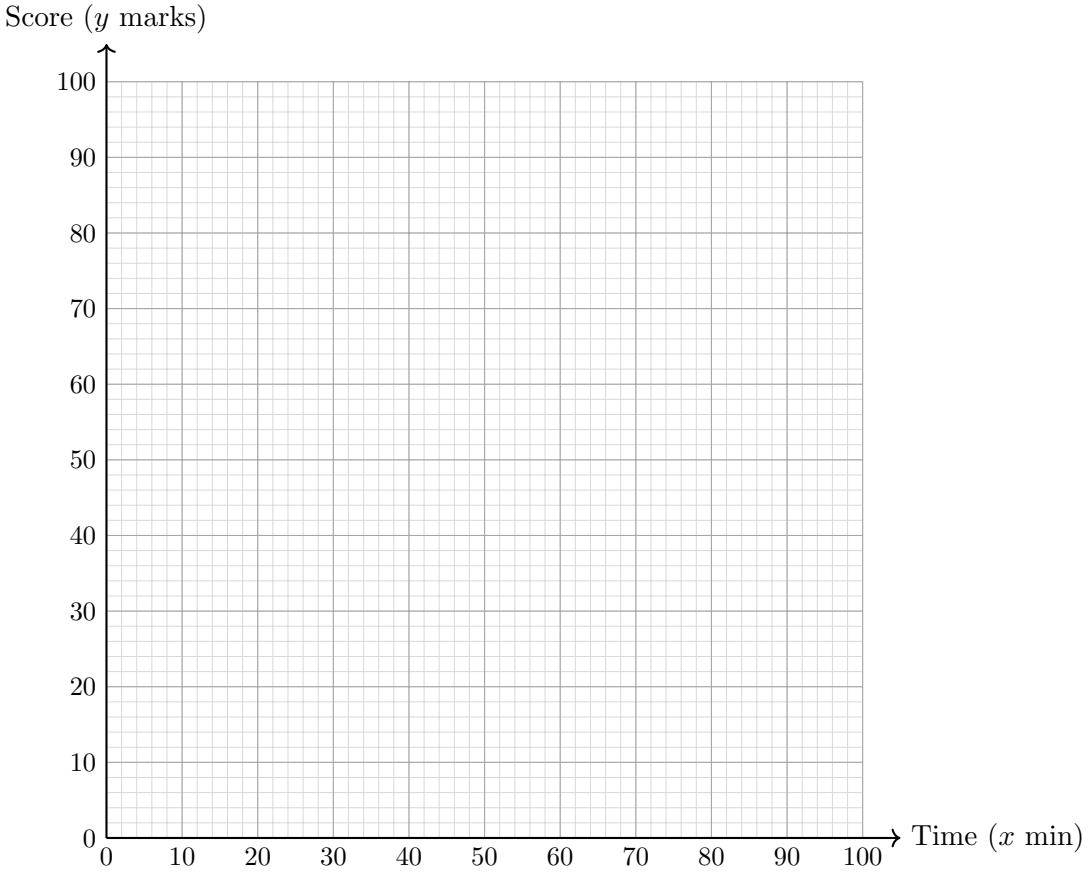
..... **[1]**

**(c)** Draw a line of best fit on the scatter diagram. **[1]**

**5** A teacher records the time, $x$ minutes, spent studying and the score, $y$ marks, for 8 students.
The table shows the results.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Time ($x$ min) | 10 | 20 | 30 | 40 | 50 | 60 | 70 | 80 |
| Score ($y$ marks) | 15 | 25 | 20 | 40 | 50 | 70 | 60 | 80 |

**(a)** On the grid, complete the scatter diagram to show this information.




**[2]**

**(b)** What type of correlation is shown by the scatter diagram?

..... **[1]**

**(c)** On the scatter diagram, draw a line of best fit. **[1]**

**(d)** Another student spent 45 minutes studying.
Use your line of best fit to estimate their score.

..... **[1]**

**6** The table shows the time spent studying, $x$ hours, and the test score, $y$ marks, for a group of 8 students.

| Study time ($x$ hours) | 2 | 3.5 | 4 | 5 | 6.5 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|
| Test score ($y$ marks) | 45 | 52 | 60 | 68 | 75 | 82 | 85 | 95 |

**(a)** Use your graphic display calculator to find the equation of the linear regression line, giving $y$ in terms of $x$.
Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**(b)** Use your equation to estimate the test score for a student who studies for 7 hours.

..... **[2]**

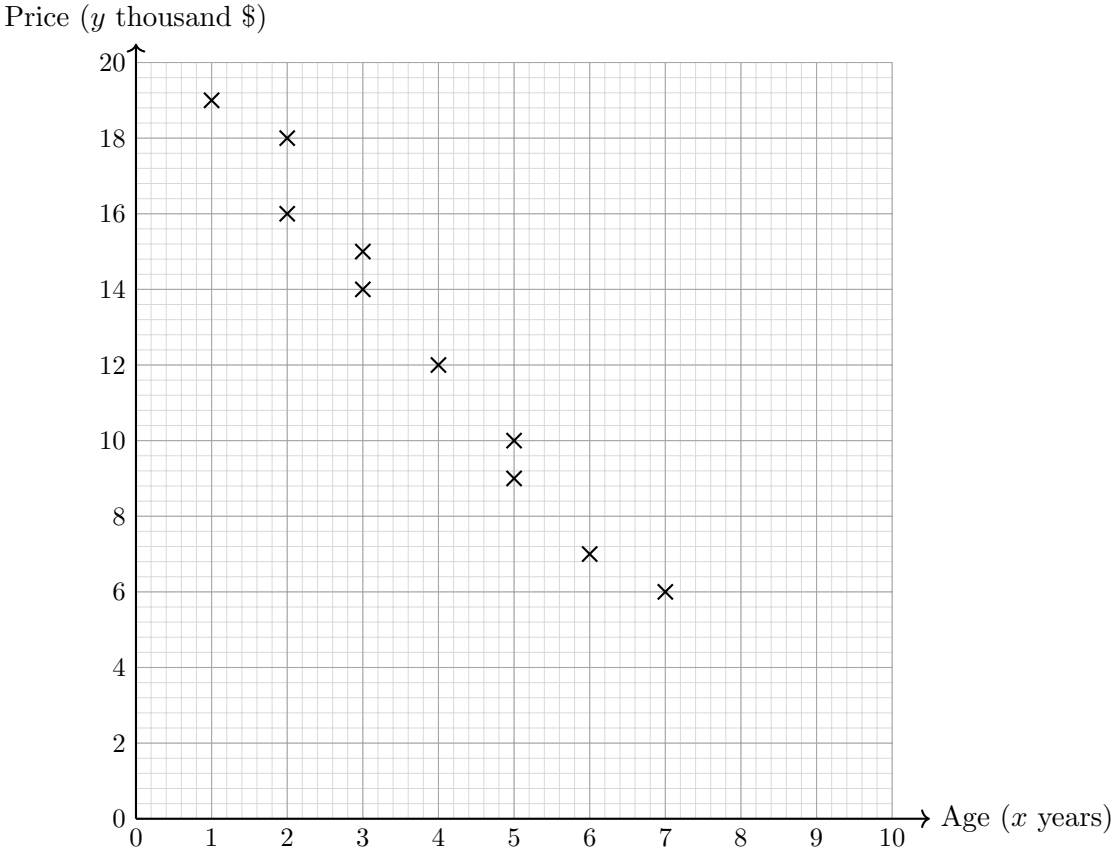
**(c)** Explain why it may not be reliable to use this equation to estimate the test score for a student who studies for 15 hours.

..... **[1]**

**7** A car dealer records the age, $x$ years, and the price, $y$ thousand dollars, of 12 used cars of the same model.
The table shows the data.

| Age ($x$ years) | 1 | 2 | 2 | 3 | 3 | 4 | 5 | 5 | 6 | 7 | 7 | 9 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Price ($y$ thousand \$) | 19 | 18 | 16 | 15 | 14 | 12 | 10 | 9 | 7 | 6 | 5 | 1 |

**(a)** The scatter diagram shows the first 10 points from the table.
Plot the remaining 2 points on the scatter diagram.




**[1]**

**(b)** Calculate the mean age and the mean price.

Mean age = ..... , Mean price = ..... **[2]**

**(c)** On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

**(d)** State the type of correlation shown.

..... **[1]**

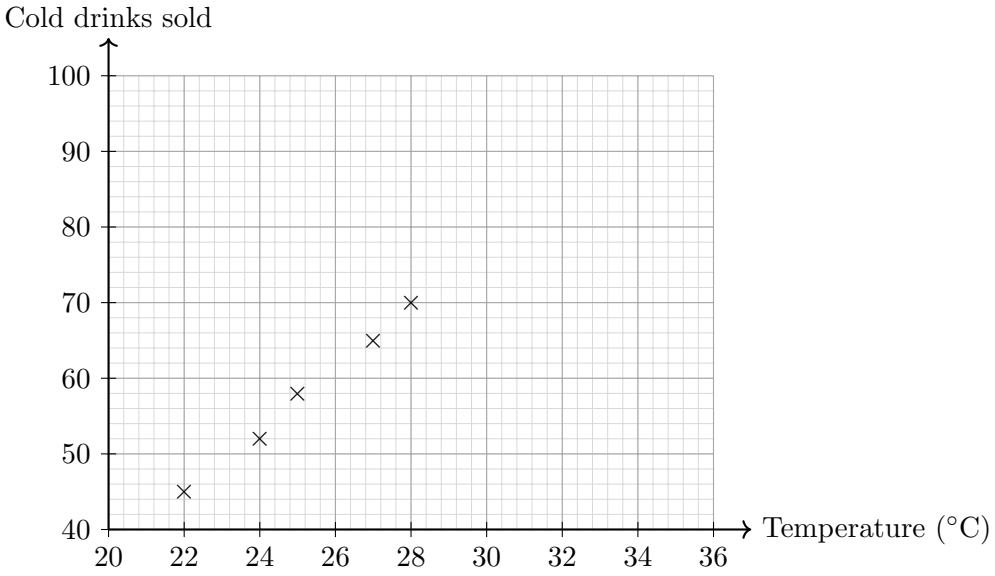
**(e)** Use your line of best fit to estimate the price of a car that is 8 years old.

..... thousand dollars **[1]**

**8** The table shows the maximum temperature, $x$°C, and the number of cold drinks sold, $y$, at a shop on 8 different days.

| Temperature, $x$ (°C) | 22 | 24 | 25 | 27 | 28 | 31 | 32 | 35 |
|---|---|---|---|---|---|---|---|---|
| Cold drinks sold, $y$ | 45 | 52 | 58 | 65 | 70 | 81 | 85 | 96 |

The first 5 points have been plotted on the scatter diagram.




**(a)** Plot the remaining 3 points on the scatter diagram. **[2]**

**(b)** Calculate the mean temperature and the mean number of cold drinks sold.

Mean temperature = ..... , Mean drinks sold = ..... **[2]**

**(c)** On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

**(d)** Predict the number of cold drinks sold on a day when the temperature is 29°C.

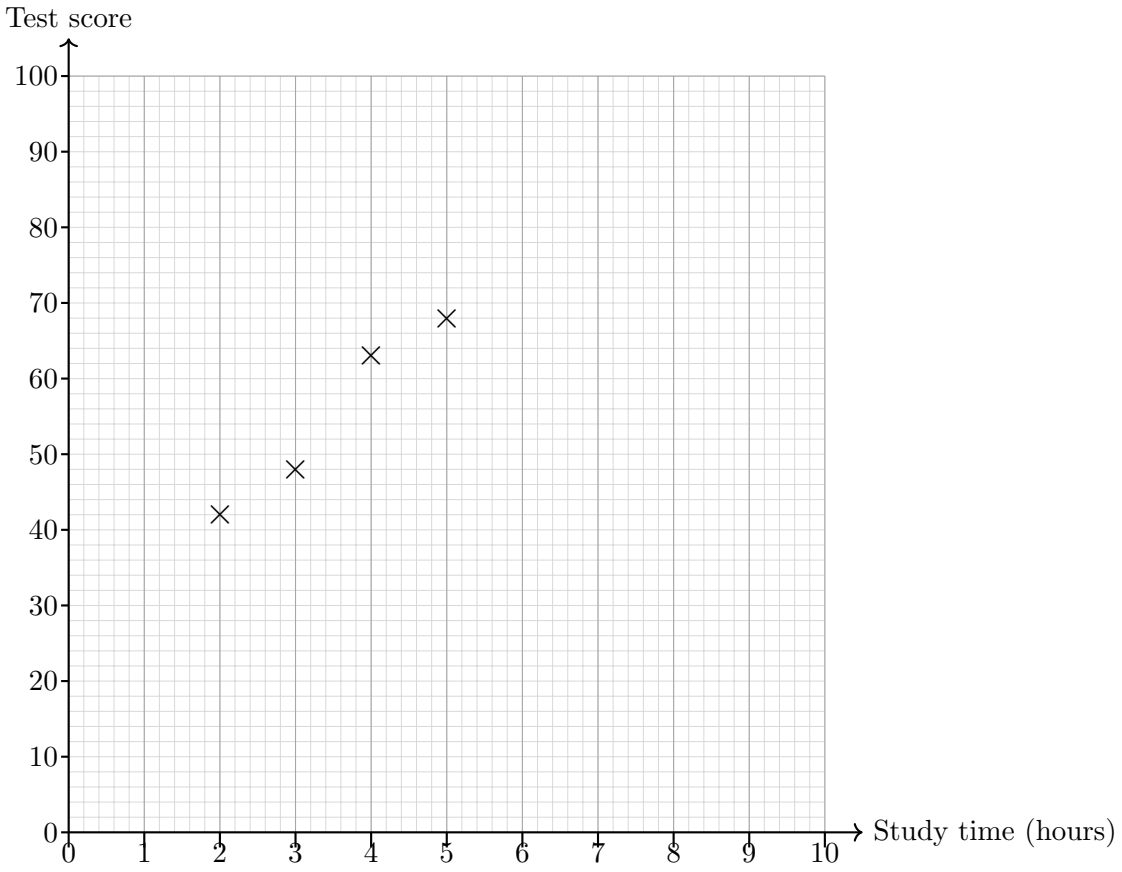
..... **[1]**

**9** A teacher records the number of hours spent studying, $x$, and the test score, $y$, for 7 students.

| Study time ($x$ hours) | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| Test score ($y$) | 42 | 48 | 63 | 68 | 81 | 88 | 100 |

The scatter diagram for this data is partially completed.

**(a)** Plot the remaining three points on the scatter diagram.




**[2]**

**(b)** Calculate the mean study time, $\bar{x}$, and the mean test score, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

**(c)** On the scatter diagram, plot the mean point and draw a line of best fit. **[2]**

**(d)** Another student studied for 5.5 hours.
Use your line of best fit to estimate their test score.

..... **[1]**

**10** A teacher records the daily revision time, $x$ minutes, and the test score, $y$ marks, for 8 students. The results are shown in the table below.

| Revision time ($x$ min) | 10 | 15 | 20 | 25 | 30 | 35 | 40 | 45 |
|---|---|---|---|---|---|---|---|---|
| Test score ($y$ marks) | 34 | 40 | 44 | 51 | 55 | 62 | 65 | 73 |

**(a) (i)** Use your graphic display calculator to find the equation of the linear regression line of $y$ on $x$, giving the coefficients correct to 3 significant figures.

$y =$ ..... **[2]**

**(ii)** Write down the coordinates of the mean point $(\bar{x}, \bar{y})$.

( ..... , ..... ) **[2]**

**(b) (i)** Use your regression equation to estimate the test score of a student who revises for 32 minutes per day.

..... marks **[2]**

**(ii)** Explain why using the regression equation to estimate the test score of a student who revises for 90 minutes per day would not be valid.

..... **[2]**

**(c)** A 9th student revises for 25 minutes per day and scores 51 marks.
State, with a reason, how adding this student's data affects the coordinates of the mean point $(\bar{x}, \bar{y})$.

.....

..... **[2]**

**11** The table shows the age, $x$ years, and the value, $y$ dollars, of 6 cars of the same model.

| Age, $x$ (years) | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| Value, $y$ (\$) | 12600 | 10800 | 9600 | 8100 | 6300 | 5100 |

**(a)** Use a graphic display calculator to find the equation of the linear regression line. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**(b)** Give an interpretation of the value of $m$ in this context.

..... **[1]**

**(c)** Use your equation to estimate the value of a car that is 4.5 years old.

% ..... **[2]**

**(d)** Explain why it may not be reliable to use this equation to estimate the value of a car that is 15 years old.

..... **[1]**

**12** A researcher measures the height, $x$ cm, and the mass, $y$ kg, of 12 adult males.
The results are summarised as follows:

$$\sum x = 2100 \qquad \sum y = 924$$

The equation of the linear regression line is $y = 0.85x - c$, where $c$ is a constant.

**(a)** Calculate the mean height, $\bar{x}$, and the mean mass, $\bar{y}$.

$\bar{x} =$ ..... , $\bar{y} =$ ..... **[2]**

**(b)** The point $(\bar{x}, \bar{y})$ lies on the linear regression line.
Find the value of $c$.

$c =$ ..... **[2]**

**(c)** A new adult male is added to the study.
His height is 180 cm.
His mass is exactly equal to the value predicted by the regression line.
Calculate the new mean mass for all 13 men.

..... kg **[2]**

**13** An automotive engineer tests 8 different cars to investigate the relationship between engine size, $x$ litres, and fuel consumption, $y$ km/l.
The results are shown in the table.

| Engine size ($x$ litres) | 1.0 | 1.2 | 1.5 | 1.8 | 2.0 | 2.4 | 2.5 | 3.0 |
|---|---|---|---|---|---|---|---|---|
| Fuel consumption ($y$ km/l) | 21.5 | 20.0 | 18.2 | 16.5 | 15.0 | 13.0 | 12.5 | 9.8 |

**(a) (i)** Use a graphic display calculator to find the equation of the linear regression line. Give your answer in the form $y = mx + c$.

$y =$ ..... **[2]**

**(ii)** Explain what the value of the gradient represents in this context.

..... **[1]**

**(b)** Use your equation from part (a)(i) to estimate the fuel consumption of a car with an engine size of 1.6 litres.

..... km/l **[2]**

**(c)** Explain why it may not be reliable to use this equation to estimate the fuel consumption of a car with an engine size of 5.0 litres.

..... **[1]**

**14** The table shows the temperature, $x$°C, and the number of ice creams sold, $y$, at a shop over 8 days.

| Temperature ($x$°C) | 12 | 15 | 18 | 20 | 22 | 25 | 28 | 30 |
|---|---|---|---|---|---|---|---|---|
| Ice creams sold ($y$) | 40 | 52 | 68 | 75 | 84 | 98 | 110 | 115 |

**(a)** Calculate the mean temperature and the mean number of ice creams sold.

Mean temperature = ..... , Mean ice creams sold = ..... **[2]**

**(b)** Use a graphic display calculator to find the equation of the linear regression line for this data. Give your answer in the form $y = mx + c$, with values of $m$ and $c$ correct to 3 significant figures.

$y =$ ..... **[2]**

**(c)** The shop owner predicts the number of ice creams sold on a day when the temperature is 24°C. Use your equation to estimate this number.

..... **[2]**

**(d)** Comment on the reliability of using your equation to predict the number of ice creams sold on a day when the temperature is 42°C.

..... **[1]**

**15** A company records its monthly advertising spend, $x$ thousand dollars, and its monthly sales, $y$ thousand dollars, over 8 months.
Two entries in the table, $p$ and $q$, are missing.
Over the 8 months, the mean advertising spend is $\bar{x} = 11$ thousand dollars and the mean monthly sales is $\bar{y} = 45$ thousand dollars.

| Advertising ($x$ thousand \$) | 4 | 7 | 8 | $p$ | 12 | 14 | 15 | 18 |
|---|---|---|---|---|---|---|---|---|
| Sales ($y$ thousand \$) | 26 | 34 | 38 | 43 | 47 | 53 | $q$ | 62 |

**(a)** Find the value of $p$ and the value of $q$.

$p =$ ..... , $q =$ ..... **[2]**

**(b)** **(i)** Use your graphic display calculator to find the equation of the linear regression line of $y$ on $x$ in the form $y = ax + b$.

$y =$ ..... **[2]**

**(ii)** Interpret the value of $a$ in the context of this question.

..... **[1]**

**(c)** Calculate the difference between the actual monthly sales when $x = 14$ and the monthly sales predicted by the regression line, giving your answer in dollars.

\$ ..... **[3]**

**(d)** **(i)** Use the regression equation to estimate the monthly advertising spend required to achieve monthly sales of \$68 000, giving your answer in thousand dollars correct to 3 significant figures.

..... thousand dollars **[2]**

**(ii)** Explain why this estimate may be unreliable.

..... **[1]**